AI-Driven Cooperative Perception and Digital Twin Platform for Connected Autonomous Vehicles (CAVs)

E. M. Malathy[1], Jami Sai Akshaya[2], Haritha M[2], Harish RP[2]
[1]Professor, [2]UG Students, B.E. CSE
Department of Computer Science and Engineering
Rajalakshmi Engineering College, Chennai, India
[1]malathy.em@rajalakshmi.edu.in, [2]230701121@rajalakshmi.edu.in, 230701107@rajalakshmi.edu.in, 230701106@rajalakshmi.edu.in

***Abstract*—Connected and autonomous vehicles depend too much on onboard sensors, so the information available to each vehicle can be limited by blind spots and occlusions, sensor effectiveness range, road and weather conditions, and inherent sensor limitations. The situational awareness of a vehicle can be reduced in complex and dynamic traffic environments. To address the situational awareness problem, in this work, we propose an AI-based platform for connected vehicles integrating multi-modal perception, cooperative information sharing, and Digital Twin technology. This connected vehicle platform derives environmental information from multiple sensing sources and shares the perception information with other connected vehicle platforms. The perception layer, based on artificial intelligence, performs object, lane, and obstacle detection. Sensor fusion merges the sensor data into a single representation of the environment, and this information is transmitted to a Digital Twin platform that represents a virtual replica of the connected vehicle and its environment. The current state of the vehicle is replicated and exhibited on the platform as an updated virtual copy that continuously evolves over time. The platform also provides services such as connected vehicle communication, clock synchronization, event detection, data storage of the Digital Twin representation, and environment visualization. A preliminary proof of concept is set up to demonstrate how the perception layer and the Digital Twin layer interact.**

***Index Terms*—Connected Autonomous Vehicles, Cooperative Perception, Digital Twin, Multi-Modal Perception, Sensor Fusion, V2V Communication, Real-Time Situational Awareness**

I. Introduction

Connected and autonomous vehicles are becoming an important part of modern transportation systems, with the ability to communicate with other vehicles, road infrastructure, and surrounding systems. These vehicles mainly depend on onboard sensors such as cameras, LiDAR, radar, and GPS to understand the driving environment and support different driving and safety-related functions. Although these sensors provide valuable information, the perception capability of a single vehicle is still limited by factors such as blind spots, occluded objects, limited sensing range, sensor noise, and changing environmental conditions. As a result, a vehicle may not always have complete information about the surrounding environment, particularly in complex and dynamic road conditions. Cooperative perception provides a way to overcome some of these limitations by allowing connected vehicles to share relevant environmental information. Instead of each vehicle independently processing only the information available through its own sensors, information from multiple vehicles can be combined to obtain a broader understanding of the traffic scene. This allows a vehicle to gain information about objects or road conditions that may not be directly visible from its own position. Previous studies have explored V2V and V2X communication for cooperative perception, where vehicles exchange information about detected objects, road conditions, and other environmental features.

Learning-based approaches and feature fusion techniques have also been investigated to improve the quality of shared perception information while reducing communication requirements. [1], [2]. Multi-modal perception might augment perception of the environment in combination with information elicited from other sensing sources. Sensing sources provide disparate information that can be combined to form a more complete dynamic traffic scene model. When talking about a connected vehicle environment, the influx of perception information from other vehicles allows augmenting local perception by combining local sensing information with reported shared information, thereby enabling the detection and identification of partly occluded objects, objects out of range of a particular vehicle's individual sensors, or objects that are difficult to identify with a single sensing modality. Digital Twin technology is another piece of connected vehicle systems. A Digital Twin is a virtual replica of the physical environment and can be continually updated with information collected from the connected vehicles. For connected and autonomous vehicles, a Digital Twin can provide information such as vehicle locations, objects detected, states of vehicles, environmental observations, or traffic events. Several recent studies have dealt with the Digital Twin structure for connected and autonomous vehicles, such as the virtual transportation environment and physical transportation environment integrated by the Digital Twin [**?**] or distributed Digital Twin architecture for the connected vehicle [**?**]. Those approaches

show the possibility of Digital Twin technology to maintain the synchronized state of the transportation environment and enable simulation, monitoring, and decision-making in near real-time. The suggested platform aims to facilitate real-time communication, vehicle-state synchronization, event processing, data storage, and visualization. To illustrate the interaction between the perception and Digital Twin layers, a proof-of-concept implementation is developed. The general purpose of the suggested approach is to enhance real-time situational awareness by extending the availability of information outside the sensing range of an individual vehicle, and constitute a basis for more reliable and intelligent CV applications.

II. Literature Survey

Connected and automated vehicles rely on multiple sensing modalities to perceive their surrounding environment. However, the perception capability of a single vehicle is constrained by sensing range, occlusion, blind spots, sensor uncertainty, and the dynamic nature of road environments. Cooperative perception addresses these limitations by allowing vehicles and infrastructure to share information through V2V, V2I, or V2X communication. Caillot et al. presented a comprehensive survey of cooperative perception in automotive applications and discussed the evolution from vehicle-only perception toward cooperative environments involving infrastructure and multiple sensing agents [1]. Early cooperative-perception research investigated the sharing and fusion of sensor information between connected vehicles. Chen et al. proposed F-Cooper, a feature-based cooperative perception framework that performs feature-level fusion of 3-D point clouds. Their study demonstrated that exchanging compact feature representations can improve object-detection performance while reducing the communication burden compared with raw sensor-data transmission [2]. The development of open datasets and simulation platforms has further accelerated research in cooperative perception. Xu et al. introduced OPV2V, an open benchmark dataset and fusion pipeline for V2V perception. The work addressed the lack of standardized large-scale datasets for evaluating cooperative perception and demonstrated the use of intermediate information fusion for improving perception under V2V communication constraints [3]. More recently, Xu et al. proposed V2X-ViT, a Vision Transformer-based framework for V2X cooperative perception. The approach performs information fusion across vehicles and infrastructure while addressing challenges including asynchronous information sharing, pose errors, and heterogeneous V2X agents. The evaluation used a large-scale V2X perception dataset generated using CARLA and OpenCDA [4]. A recent survey of V2X cooperative perception further identifies agent selection, data alignment, feature fusion, uncertainty, communication delay, and data loss as important challenges for practical cooperative-perception systems [5]. These observations indicate that cooperative perception is not only a perception problem but also a communication and distributed-systems problem. Digital Twin technology provides a complementary approach for representing and monitoring physical transportation systems through continuously updated virtual models. In the context of connected and automated vehicles, a Digital Twin can represent vehicles, infrastructure, traffic conditions, sensing information, and communication states. The Digital Twin concept therefore provides a mechanism for connecting physical or simulated transportation entities with their virtual counterparts. Wang et al. investigated Digital Twin simulation for connected and automated vehicles using the Unity game engine. Their architecture separated the physical and digital environments and demonstrated the use of Digital Twin simulation for connected and automated-vehicle applications [6].

The role of Digital Twins specifically in connected and automated vehicles was examined by the authors of The Role of Digital Twins in Connected and Automated Vehicles. The study emphasized that a Digital Twin requires communication between the physical system and its digital model and discussed its relevance to CAV development, testing, and simulation [7]. Wang et al. also investigated automatic traffic modelling for creating Digital Twins for autonomous-vehicle development. Their work highlights the importance of traffic simulation and digital representations for testing autonomous-driving systems under region-specific and dynamic traffic conditions [8]. More recently, Ge and Qin conducted a systematic review of Digital Twin Intelligent Transportation Systems (DT-ITS). Their review identifies architecture design, data synchronization, real-time analysis, connectivity, and implementation challenges as important research issues. The study also highlights the increasing interaction between Digital Twins, IoV, V2V/V2I communication, and cooperative perception [9]. A recent comprehensive survey by Gu et al. further examined Digital Twin applications in intelligent vehicles and transportation systems, covering vehicle safety, traffic management, autonomous driving, simulation, testing, and real-time data analytics. The survey identifies Digital Twins as an important technology for integrating physical transportation systems with virtual models and highlights open challenges and future research directions [10]. Simulation plays an important role in the development and evaluation of connected and autonomous vehicle technologies because large-scale real-world experimentation can be expensive, time-consuming, and difficult to reproduce. CARLA is an open-source urban-driving simulator developed specifically for autonomous-driving research and provides simulated urban environments, vehicles, sensors, and configurable traffic scenarios [11]. CARLA has subsequently been used in cooperative-perception research to generate multi-agent traffic scenarios and datasets. For example, the V2X-ViT study used CARLA together with OpenCDA to construct a large-scale V2X perception dataset for evaluating cooperative 3-D object detection [4]. This demonstrates the suitability of simulation environments for investigating the interaction between perception, communication, and multiple connected agents.The use of simulation is also particularly relevant to Digital Twin research because the simulated environment can act as a controllable representation of the physical transportation environment. Consequently, a simulation-based Digital Twin platform can provide an intermediate step be-

tween algorithm development and real-world deployment. The above literature demonstrates significant progress in cooperative perception and Digital Twin technology. However, these two research directions have largely evolved through different research objectives. Cooperative-perception studies primarily concentrate on improving environmental perception through information exchange, data fusion, and AI-based detection. Digital Twin studies, in contrast, focus on maintaining a virtual representation of physical entities, simulation, monitoring, prediction, and transportation-system management. Recent Digital Twin reviews indicate increasing interest in combining Digital Twins with IoV, V2V/V2I communication, and cooperative perception [9], [10]. Nevertheless, there remains scope for implementation-oriented platforms that demonstrate the complete pipeline This motivates the present work. Rather than proposing another isolated perception algorithm, this study develops a simulation-based end-to-end platform in which structured vehicle and perception events are generated, communicated through an MQTT-based mechanism, processed by a backend service, and used to update the corresponding Digital Twin. The current implementation is intended as a proof-of-concept and provides a foundation for subsequent integration of AI-based object detection, multimodal sensor fusion, communication-performance evaluation, and large-scale connected-vehicle experiments.

A. Research Gap

Based on the reviewed literature, the following research gaps are identified: Cooperative perception and Digital Twin technologies are often studied independently. Existing cooperative-perception studies mainly emphasize detection accuracy and information-fusion algorithms, whereas an end-to-end Digital Twin synchronization pipeline is less emphasized. Digital Twin studies frequently concentrate on modelling, monitoring, simulation, and transportation management, while detailed vehicle-level cooperative perception and event exchange are comparatively less central. Communication constraints such as delay, packet loss, asynchronous information, and data consistency remain important challenges in cooperative perception [5]. There is a need for reproducible simulation-based platforms that connect vehicle perception, communication, backend processing, and Digital Twin synchronization. Accordingly, the present work focuses on the implementation of an integrated platform rather than claiming a new perception algorithm. The main novelty lies in the system-level integration and proof-of-concept demonstration of structured cooperative information exchange and Digital Twin updating in a connected-vehicle simulation environment.

B. Novelty

The contribution of the proposed work is the integration of cooperative information sharing and Digital Twin synchronization within a common connected-vehicle architecture. Instead of transmitting raw sensor streams, the prototype represents observations as structured events containing vehicle, object, position, velocity, timestamp, confidence, and related information. These events are exchanged through the communication layer and processed by a backend service before updating the Digital Twin. The present work demonstrates this end-to-end pipeline in simulation; advanced AI perception and multimodal sensor fusion are identified as extensions rather than experimentally validated components of the current prototype.

C. Objectives

The main objective is to develop and demonstrate a connected-vehicle platform that combines cooperative information sharing with Digital Twin technology for improved situational awareness. The specific objectives are to: (i) generate vehicle and environmental information in a simulated road network; (ii) represent observations using structured perception events; (iii) exchange telemetry and event information between connected vehicles and the backend through MQTT; (iv) maintain a synchronized Digital Twin state; and (v) provide a monitoring interface for vehicle states and detected events. The proposed system is evaluated as a proof-of-concept. The current evaluation focuses on the interaction among the simulation, communication, backend, and Digital Twin layers. Quantitative evaluation of AI-based detection accuracy, multimodal sensor fusion, scalability, and real-world communication performance is outside the scope of the present implementation.

III. Proposed System

The proposed system integrates a connected-vehicle simulation environment with cooperative information sharing and a Digital Twin. The architecture maintains a synchronized representation of connected vehicles and their surrounding environment. The perception layer can consume information from vehicle sensors and represent relevant observations as structured events containing attributes such as source vehicle, position, velocity, confidence, timestamp, and sensor source. The structured perception information produced by each vehicle is transmitted through the communication layer. A receiving vehicle or backend service can combine shared observations with locally available information to obtain a broader view of the environment. The Digital Twin consumes validated vehicle-state and perception information and maintains the corresponding virtual representation, including vehicle positions, speeds, detected objects, event timestamps, and synchronization status. For the present implementation, the system is developed and tested in a simulated environment. The road network is generated using OpenStreetMap data and simulated using the CARLA platform. Two simulated buses are used to demonstrate vehicle telemetry, event communication, and Digital Twin synchronization. A backend service receives the events, validates the data, updates the Digital Twin state, and provides information for visualization and monitoring. The overall objective is to demonstrate an end-to-end data flow from connected-vehicle simulation and structured information sharing to Digital Twin synchronization. The prototype provides a foundation for future extensions involving

AI-based object detection, multimodal sensor fusion, additional vehicles, communication impairments, and quantitative cooperative-perception evaluation. The overall architecture of the proposed system is shown in Fig. 1.

IV. System Architecture

The proposed system follows a layered architecture that connects vehicle perception, cooperation and information sharing, communication services, backend processing, data storage, analytics, and the Digital Twin. The architecture enables structured information collected from physical or simulated vehicles to be transmitted, processed, and represented through their corresponding digital counterparts.

A. Simulation and Scenario Layer

This layer provides the simulated urban road environment and connected vehicle scenarios. It represents the road network, connected buses, surrounding traffic, and different traffic events required for testing the proposed system.

B. Telemetry and Communication Layer

This layer collects vehicle state and event information from the simulated buses and enables communication between the connected vehicles and the backend system. MQTT is used as the communication mechanism for exchanging telemetry and control information.

C. Backend Application Layer

The backend layer receives and processes the information generated by the connected vehicles. It performs message validation, Digital Twin state management, synchronization, freshness monitoring, reliability analysis, event processing, and situational-awareness functions.

D. Data Persistence Layer

This layer stores vehicle information, vehicle states, Digital Twin states, events, communication records, synchronization records, and experiment-related data. The stored information can be used for monitoring, analysis, and evaluation of the system.

E. Analytics and Metrics Layer

The analytics layer evaluates the collected data using different performance and reliability metrics. It supports the analysis of vehicle states, communication performance, synchronization behaviour, event propagation, and other experimental measurements.

F. Experiment and Configuration Layer

This layer manages the configuration and execution of different simulation scenarios. It supports the definition of scenario conditions, network conditions, synchronization settings, events, and experiment parameters, followed by the collection of results.

G. Dashboard and Client Layer

The dashboard layer provides a visual interface for monitoring the connected buses and their Digital Twin representation. It can display vehicle positions, vehicle states, events, alerts, communication status, and other relevant system information.

H. Perception Layer

Perception layer: This layer gathers data on the vehicle and its environment. The sensing sources used (camera, LiDAR, radar, GPS, or any other available) depend on the simulation or dataset chosen. Information is extracted by computer vision and machine learning techniques for relevant objects and events. Perception events are produced in a structured form. In each perception event, these pieces of information are stored: source vehicle, timestamp, object or event detected, object identified, object position, object velocity (if any), confidence value, sensor source, coordinate frame, and event identification. This format permits a common interpretation of the information between communication and Digital Twin components.

I. Cooperative Perception Layer

For the prototype, we consider sharing object-level information instead of raw sensor data or the large intermediate feature maps. This greatly reduces the communication bandwidth, and it is also a practical approach for the cooperative perception prototype in our system.

J. Communication and Integration Layer

An integration layer: Gets received events, checks its syntax, directs them to appropriate services, and solves communication issues (duplicated events, wrong payload data, timestamp jumped, temporary network problem, etc.)

K. Digital Twin Layer

Digital Twin Layer: The Digital Twin layer is a virtual copy of the connected vehicles and their environment. It takes in the structured perception and vehicle-state information and updates the virtual state because of this. The Digital Twin stores the state of the vehicle ID, position, relative speed, objects that are detected, any subsequent event with corresponding timestamps, Synchronized state, etc.

As the platform continuously updates this data, the same information can be used to provide a global virtual building block.

L. Data and Monitoring Layer

The data and monitoring layer captures vehicle states, structured perception events, communication records, Digital Twin states, and other experiment information for live monitoring and later analysis. A backend service manages the data flow between the communication layer and the Digital Twin. The monitoring interface displays vehicle positions, states, events, alerts, communication status, and synchronization information.

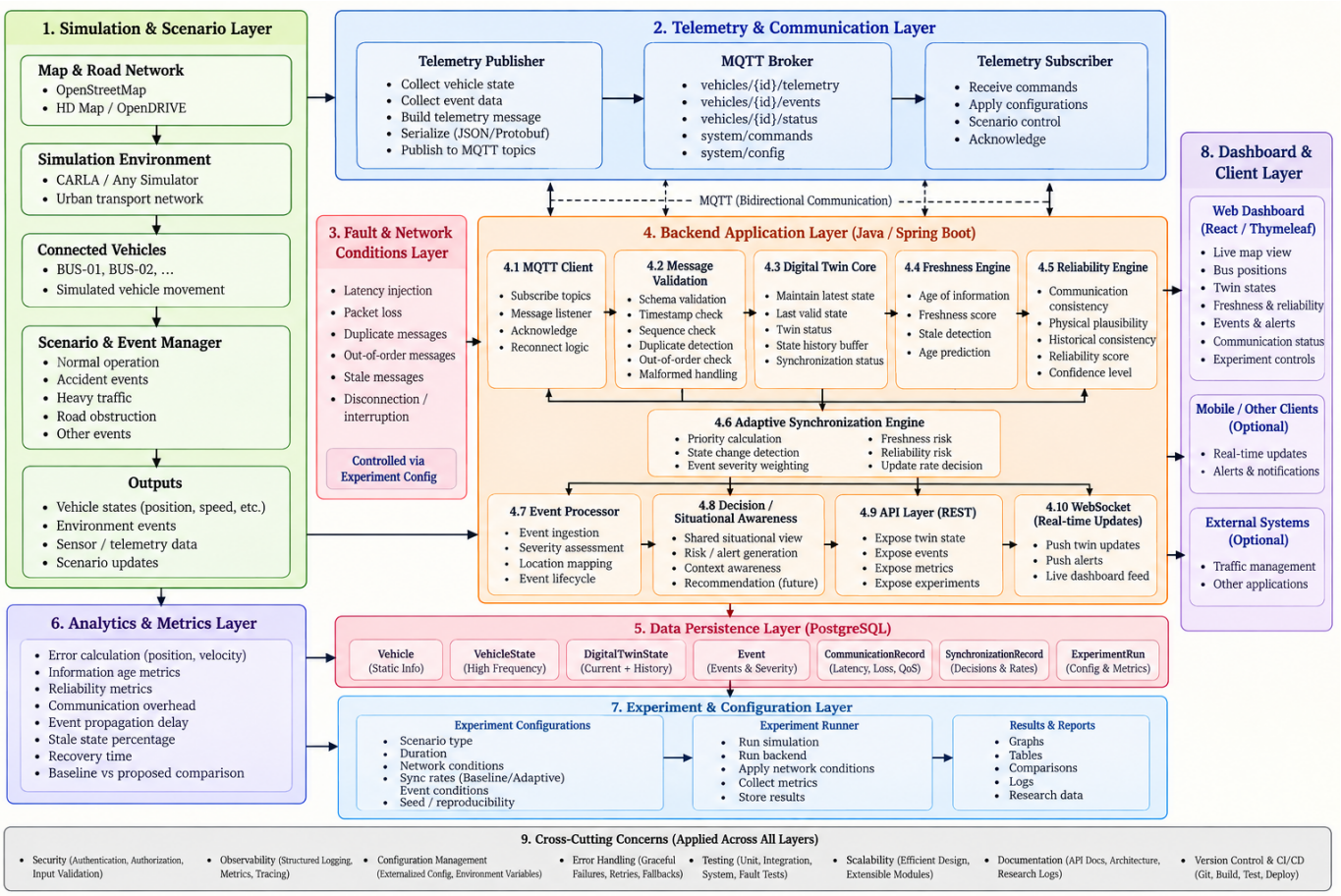


Fig. 1. System Architecture of the Proposed Connected Vehicle Digital Twin Platform

V. Methodology

The summarized system is designed as a comprehensive pipeline that integrates vehicle perception, cooperative information sharing, communication module, and Digital Twin updating. The approach is characterized by a step-by-step procedure running from a simulated transportation environment generation to perception, information sharing, Digital Twin updating, as well as system assessment.

A. Simulation Environment

The proposed workflow integrates vehicle simulation, structured perception information, cooperative information sharing, communication, backend processing, and Digital Twin updating. The procedure proceeds from transportation-environment generation through vehicle-state and event generation, information exchange, Digital Twin synchronization, and system monitoring. OpenStreetMap data are used to generate the urban road environment, which is then configured in the CARLA open-source simulator. The present proof-of-concept uses two simulated buses so that vehicle-state exchange and cooperative event communication can be observed clearly

B. Vehicle Perception

C. Vehicle Perception

The perception module uses information from the simulated vehicle sensors to identify relevant objects and environmental conditions. In the current proof-of-concept, the focus is on converting these observations and vehicle events into a structured format for communication rather than evaluating AI detection performance. Each perception event includes details such as the source vehicle, timestamp, object ID, object class, position, velocity, confidence, sensor source, coordinate frame, and event ID. This provides a common interface between the perception and communication modules.

D. Sensor Fusion

When multiple sensing modalities are available, their observations can be combined in a common coordinate frame and synchronized over time to obtain a more consistent representation of the environment. The current proof-of-concept architecture supports this extension, but a quantitative multimodal sensor-fusion experiment is not included in the present results.

E. Cooperative Information Sharing

The cooperative perception stage enables one connected vehicle to share relevant observations with another vehicle or the backend platform. For the initial prototype, object- and event-level information is exchanged instead of raw sensor data or large intermediate feature representations. This approach reduces communication requirements and provides a practical basis for evaluating cooperative information sharing. Future quantitative evaluation can compare local-only and cooperative configurations using detection quality, message size, update frequency, latency, event delivery ratio, and handling of missing information. Such comparative measurements are not reported in the present proof-of-concept results.

F. Communication and Data Integration

Incoming structured events pass through the communication layer using MQTT-based publish–subscribe exchange. REST-based interfaces can be used for service access and data retrieval. A backend service receives each event, validates its structure, and forwards the validated information to the appropriate Digital Twin component. The integration process also considers distributed-system conditions such as duplicate events, invalid messages, timestamp inconsistencies, network connection failures, and system failures. These conditions are handled to maintain consistency between the received information and the Digital Twin state.

G. Digital Twin Synchronization

The Digital Twin accepts validated vehicle and perception events and updates the corresponding virtual vehicle state. The virtual state contains vehicle identity, position, velocity, object information, event timestamps, and synchronization status. virtual state contains the identity, position, and velocity of the vehicle and objects, the perception event timestamp, and the synchronization state. The received data and information are used to update the state of the relevant vehicle and environment by the Digital Twin. This results in an always up-to-date digital replica of the connected vehicles and gives a uniform representation for access.

H. Monitoring and Evaluation

The synchronized Digital Twin state is accessible to users using a monitoring interface. Three screens are proposed: the dashboard, which shows the current states of both vehicles, shared events, and synchronization information. The system is tested at the perception and system levels. Perception performance can be evaluated based on various tools like precision and recall, mAP, and, if applicable, IoU. Cooperative perception performance can be analyzed by comparing local-only and cooperative cases. For the system level, one can consider the end-to-end la-tency, message throughput, event delivery ratio, state freshness of the Digital Twin, reliability of the communication, recovery from the communication failure, and so on, to study whether this pipeline can provide a robust and timely digital twin of the connected vehicles.

VI. Results and Discussion

The proposed system was implemented as a proof-of-concept in a simulated connected-vehicle environment. The experiment brings together the CARLA simulation, connected buses, MQTT-based communication, backend processing, and Digital Twin monitoring as a single workflow. The buses generate vehicle telemetry and event information during the simulation, which is transmitted through the communication layer and processed by the backend before being reflected in the Digital Twin. This setup was used to verify the basic end-to-end operation of the proposed framework. The current evaluation focuses on demonstrating the system workflow and initial communication behaviour, while detailed AI perception and large-scale scalability evaluation are planned for the next stage of development.

A. Simulation Output

Figure 2 The simulation output shows the connected buses operating within the CARLA road environment. During the simulation, the vehicles move through the defined road scenario while generating telemetry and event information. These observations are passed to the communication and backend layers for further processing. The simulation therefore provides the basic environment required to test the interaction between the connected vehicles and the Digital Twin.

B. Event Detection and Communication

The system was tested for the exchange of vehicle telemetry and event information between the connected buses. During the simulation, when an event occurs in one vehicle, the relevant information is collected and sent through the communication layer to the backend. The backend processes the received message and updates the corresponding vehicle state in the Digital Twin. The processed information can then be shared with the other connected vehicle when required. This experiment demonstrates how an event generated by one vehicle can be communicated and reflected across the connected-vehicle system, forming the basic cooperative information-sharing workflow.

C. Digital Twin Monitoring

The telemetry and event information received from the connected vehicles are used to continuously update their corresponding Digital Twin states. The Digital Twin maintains the latest information about each vehicle, including its position, movement, and detected events. This information is displayed through the monitoring interface, allowing changes in the simulated environment to be observed in near real time. The experiment shows the vehicle states and events are successfully reflected in the Digital Twin, providing a synchronized view of the connected vehicles and supporting overall system monitoring.

graphicx



Fig. 1. CARLA simulation environment showing the urban road scenario with connected buses.



Fig. 2. Close-up view of the connected bus in the CARLA simulation environment.



Fig. 3. Telemetry messages and accident event detected during simulation.



Fig. 4. Continuous telemetry data from connected buses during simulation.

Fig. 2. Simulation output of the proposed connected vehicle system.

D. Telemetry Freshness Analysis

Telemetry freshness was measured to see how closely the Digital Twin follows the current state of the connected vehicles. The measurements were collected from the telemetry messages exchanged between the simulated buses and the backend system. Figure 3 shows the freshness values recorded during the experimental runs.

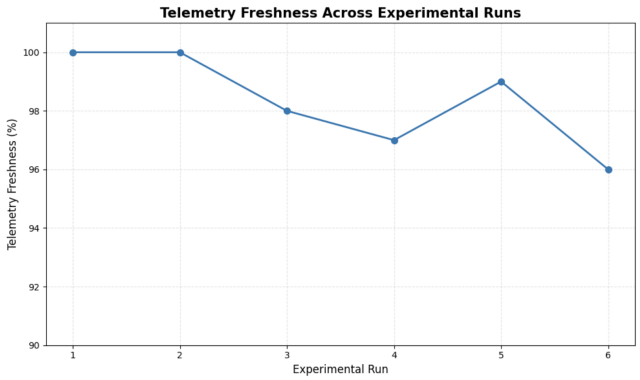


Fig. 3. Telemetry freshness across experimental runs.

The results show that the system maintained high telemetry freshness throughout the runs. This indicates that the Digital Twin received recent vehicle-state updates and was able to keep the vehicle representation up to date. Consistent freshness also helps the system maintain a reliable view of the connected vehicles during event processing. This is particularly important when the vehicle state changes frequently during the simulation.

E. Event Propagation Latency

Event propagation latency was measured to determine the time-taken for an event generated by one connected vehicle to reach and be processed by the backend. Figure 4 shows the latency observed during the experimental runs.

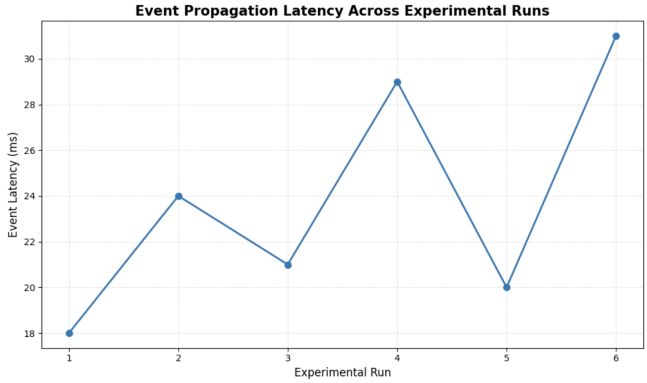


Fig. 4. Event propagation latency across experimental runs.

The results show the variation in event propagation time between the different experimental runs. The observed latency gives an initial indication of how quickly the system can receive and process events generated by the connected vehicles. Lower latency means that the event information can reach the backend with less delay, which is useful for maintaining timely updates in the Digital Twin. These measurements also provide a basis for further evaluation of the communication and processing performance as the system is extended.

F. Event Delivery Performance

Event delivery performance was evaluated to check whether the events generated by the connected vehicles were successfully received by the backend. Figure 5 shows the event delivery rate observed during the experimental runs.

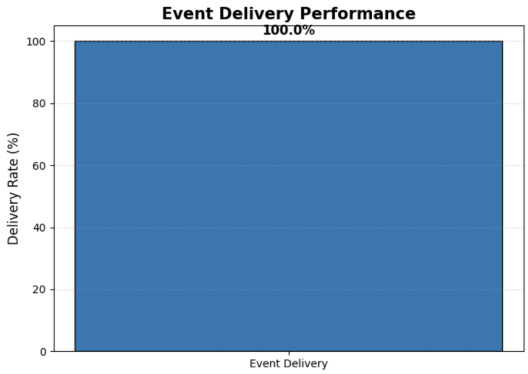


Fig. 5. Event delivery performance of the proposed Digital Twin framework.

The results indicate that the generated events were successfully transmitted through the communication pipeline. This confirms that the backend was able to receive the event information required to update the Digital Twin and maintain the current system state.

VII. Conclusion

The proposed system presents a connected vehicle framework that integrates cooperative perception, vehicle-to-vehicle communication and Digital Twin technology. The proof-of-concept developed was deployed in a simulated environment with connected buses. Vehicle telemetry and event information was communicated through the communication layer and processed by the backend. The received information was then used to keep updated the corresponding vehicle states in the Digital Twin.

The experimental results demonstrate that the basic end-to-end functionality of the system, including the exchange of vehicle telemetry, transmission of events, backend processing and monitoring of the Digital Twin, works. The system also provides an initial view on the communication behaviour via parameters such as telemetry freshness, event propagation latency, and event delivery. The results demonstrate that the proposed framework can be used as a basis for maintaining shared vehicle information and improving situational awareness in a connected environment.

The current implementation is primarily meant to verify the overall system workflow in simulation. Future works will focus on the integration and evaluation of AI-based perception models, the use of more realistic sensor data, the increase of the number of connected vehicles, and testing of the framework in larger and more complex road scenarios. These extensions can support the advancement of the current proof-of-concept toward a more complete cooperative perception and Digital Twin platform for connected autonomous vehicles.

VIII. Future Work

The current prototype will be extended in future work to support multiple connected vehicles and more complex urban traffic scenarios. The perception module could be advanced by integrating AI-based object and obstacle detection, and its performance can be evaluated using metrics like precision, recall, mAP, and IoU. The system can be extended to support multimodal sensor fusion by fusing data from different vehicle sensors. This can be helpful in improving the quality of shared perception information and providing a better understanding of the surroundings.

Further experiments can compare local perception with cooperative perception, and study the communication latency, message reliability and Digital Twin freshness under different conditions. Fault recovery and handling communication failures can also be investigated to improve the robustness of the system. Also, real-world vehicle data can be input to research the behaviour of the framework beyond simulation. The proposed approach can be further assessed in more realistic and complex connected-vehicle scenarios by exploring larger DT environments and traffic decision-support mechanisms.

References

[1] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, "Opv2v: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication," in *2022 IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[2] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, "V2x-vit: Vehicle-to-everything cooperative perception with vision transformer," in *Computer Vision – ECCV 2022*, 2022, pp. 107–124.

[3] A. Caillot, S. Ouerghi, P. Vasseur, R. Boutteau, and Y. Dupuis, "Survey on cooperative perception in an automotive context," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 9, pp. 14 204–14 223, 2022.

[4] Q. Chen, X. Ma, S. Tang, J. Guo, Q. Yang, and S. Fu, "F-cooper: Feature based cooperative perception for autonomous vehicle edge computing system using 3d point clouds," in *Proceedings of the 4th ACM/IEEE Symposium on Edge Computing*, 2019, pp. 88–100.

[5] R. Xu *et al.*, "Vehicle-to-everything cooperative perception for autonomous driving," *Proceedings of the IEEE*, vol. 113, no. 5, pp. 443–477, 2025.

[6] Z. Wang, K. Han, P. Tiwari *et al.*, "Digital twin simulation of connected and automated vehicles with the unity game engine," in *2021 IEEE International Conference on Digital Twins and Parallel Intelligence*, 2021.

[7] S.-H. Wang, C.-H. Tu, and J.-C. Juang, "Automatic traffic modelling for creating digital twins to facilitate autonomous vehicle development," *Connection Science*, vol. 34, pp. 1018–1037, 2022.

[8] C. Ge and S. Qin, "Digital twin intelligent transportation system (dt-its)–a systematic review," *IET Intelligent Transport Systems*, 2024.

[9] X. Gu, W. Duan, G. Zhang, J. Hou, L. Peng, M. Wen, F. Gao, M. Chen, and P. H. Ho, "Digital twin technology for intelligent vehicles and transportation systems: A survey on applications, challenges and future directions," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 3235–3271, 2026.

[10] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "Carla: An open urban driving simulator," in *Proceedings of the 1st Annual Conference on Robot Learning*, vol. 78, 2017, pp. 1–16.